Sia $f: \mathbb{R}^3 \to \mathbb{R}^3$ l'endomorfismo

associato $M_{EE}(f) \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 4 \\ 0 & 1 & -1 \end{pmatrix}$ rispetto alla base canonica E

a) $f(x,y,z) = (x+2y,\ 2x+4z,\ y-z)$

b) **dim e base** $\ker(f)$ e $\mathrm{Im}(f)$ $\quad \begin{pmatrix} 3 & 6 & 0 \\ 5 & 2 & 2 \\ 2 & 4 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & -8 & 2 \\ 0 & 0 & 0 \end{pmatrix}$ $\quad \dim(\mathrm{Im}(f)) = 2 \to \{(1,5,2),(5,2,2)\}$

$\ker$

$A\underline{x} = 0 \to \begin{cases} x+2y=0 \\ -8y+2z=0 \end{cases}$

$\begin{cases} x=-2y \\ z=4y \\ y=y \end{cases} \to \{(-2,1,4)\}$ Banale

$\dim(\ker) = 1$

Sia $\phi_\lambda : \mathbb{R}^3 \to \mathbb{R}^3$

$\phi_\lambda(x,y,z) = (x+\lambda y,\ y+\lambda z,\ z+\lambda x)$

esiste $\lambda \in \mathbb{R}$ che ϕ_λ → suriettiva?

↳ per ogni $w \in W$ esiste $v \in V$ t.c. $\phi_\lambda(v) = w$?

Sia $V = \mathbb{R}^3$

$v_1 = (1,2,1)$

$v_2 = (0,1,1)$

$v_3 = (1,0,-1)$

$B = \{v_1, v_2, v_3\}$ è una base di $\mathbb{R}^3$

$n = 3$

$\det \begin{pmatrix} 1 & 4 & 1 & 2 \\ 2 & 3 & 5 & 0 \\ 4 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} = 2 \cdot 1 \cdot \det \begin{pmatrix} 1 & 2 \\ 5 & 0 \end{pmatrix} = -20$

$f \circ f: \mathbb{R}^3 \to \mathbb{R}^3$

$A \cdot \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \qquad v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

↳ $f \circ f \to f(f(v))$

↳ $f(v) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$

$f(f(v))$

$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$

$\begin{pmatrix} 3 & 6 & 0 \\ 5 & 2 & 2 \\ 2 & 4 & 0 \end{pmatrix}$